

ActInf GuestStream 075.1 ~ Matsumura et al.: "Active Inference With Empa

GuestStream | Matsumura et al. | 47:48

hello and welcome it's March 13th 2024
and we're here in active inference guest
stream number

75.1 active inference with empathy
mechanism for socially behaved
artificial agent in diverse situations
with t Yuki matsumura kanako isaki and
huki Mizuno so thank you all for joining
and looking forward to hearing your talk
and

discussion okay thank you for inviting
us to such a good opportunity and I'm T
and my co researcher kanako and huki
also join this
discussion we are developing AI
especially AI which is inspired by
humanik autonomous agent to make social
artificial agent recently we published a
paper related to the idea in Journal of
a life the title is active imp with
empathy mechanism for socially behaved
artificial agent in diverse
situations first I'd like to explain the
paper this is the motivation of This
research at the title of the paper shows
the motivation of This research is to
develop socially behaved artificial
agent we believed that sociality will
become an important point to apply AI in
our daily
life the difficulty to realize social
agent is that an appropriate social
behavior depend on a given situation or
culture as described in the bottom
figure we have sociality even in just

working case for example how fast should we walk or how much distance from others orah blah blah so if we Implement sociality with rule based program we have to design a lot of magic number for each situations this is impossible because there are a white variety of situations due to this unified mechanism or principle for social behavior is required this is the challenge we assume that we humans has such unified behavior model because we humans can behave socially in diverse situations and we develop social Eng inspired by human behavior model namely three principle three energy principle and active influence in this research we use the active influence for the basis of the behavior model of human life agent as you know active inference is a concept proposed in the free energy principle and free energy principle is a unified hypo for our variety of cognitive activities in the FP our cognitive activities are explained by such that first we have internal models for predicting future of surrounding environment and second we always try to minimize the free energy of the model free energy is defined by this equation and it intuitively represent uncertainty of prediction the interesting point of the FP is that the internal model if not only the hidden state but also actions

for the agent that is the action to be taken is the action which minimize the free energy of the agent this is the basic idea of active inference as I understand

it intentional actions also can be explained in the active inference in this case we consider expected free energy which is the free energy for the future State when considering much time Step Ahead expected energy at much time Step Ahead is considered as described in the right

figure based on the expected free energy for each action the distribution the action is determined such that the smaller the expected free energy the higher the probability of

section when we consider intentional Behavior intentions are considered as PRS of future observations and reward according to the Future observation are encoded into the first time into the uh in the last

equation these are my understanding of the standard active inference and the behavior model of the humanik agent based on

it we extend this active inference to generate the social behavior the idea of the extension is explained using a situation in which there are two agents called I and other the agent called I is the subject of the action in this explanation

the biological agent under the AC inference makes predictions about the environment and take some action to

minimize the uncertainty of the prediction in this figure I predict future beh I predict future behavior of the other because the a is an environment for I namely I act to reduce uncertainty for the for the other so far this is a standard active inference here if we consider that the other is also a biological agent we can assume that the other also act under the active inference under this idea we can assume that the other also predict the future of behavior of me and the other tr to is the uncertainty of the prediction about me in this session in this understanding we notice that our action can reduce not only our own free energy but also that of the other this is because my action affects not only my own free energy but also the free energy of the others if I can predict others prediction for me I can act to I can act to release the free energy of the other this is the idea of the proposed extension of the active inference we extend FP and active inference from an individual idea to an Collective idea as described in this image of concept the agent imaginary share the brain and they try to minimize the free energy of the shared brain to relas the free energy for the shared brain we act not only for me but also for the others because of this action for others we expect that this idea can be used to generate social behavior which is a behavior for us not only only for me or only for

you to realize this idea it is necessary to predict others prediction about myself this is implemented based on the mirror system and simulation Theory hypothesis for the mechanism of empathy in this in the mirror system and simulation Theory it is that that we understand others by simulating others by own bodies we consider the idea of using our own body to simulate others as using our internal model to infer other prediction the overview of the mechanism is described in the figure the agent has the internal model to predict future observation for his own observation moreover we use the same internal model to predict others prediction about myself At the model is the same the input to the model is changed to the observation of the others for this the process of inference of others observation is included for example when the observ is an image Noel view synthesis which is the problem of generating images from different Viewpoint can be used in this in the other case as we will discuss in the following evaluation if the observation is the location of the agent for ordinate transformation transformation can be used based on this inference procedures the expected free energy is is expended as described in this equation as with the encoding of intentional Behavior we add a time to the first time that is intentional to

respond to the expected expectation of others if there are multiple others the same of each other expectation is considered we call the proposed idea empathic active

we evaluate the proposed idea using the control task of an autonomous autonomous mobile agent in the evaluation multiple agent move to each Target point the shortest path of each agent intersect each other and agent are required to take collision avoidance behavior the player this red agent is controlled by the proposed method and the other agent are controlled by the social force model social force model is a simple social working model roughly set it is a Dynamics model in which the walking speed and direction are determined by two

forces the first one is driving force to the go and the second is repulsive force from the others in this evaluation the weight of the weight the weight of these two type of forces are changed to evaluate different situations where are two type of others selfish or altruistic others when the force to the goal is larger than the force from others the agent take behavior in selfish on the other hand the force from the others is larger than the PO to the go the agent take behavior in altruistic this is an example of the behavior of others under the control of the social force model the player let CLE randomly moved and the others green blue or orange moved with social force model

the top picture is an example for selfish others and the bottom picture is an example for an aristic
a the difference is from the difference of the magnitude of the two forces as you can see from the figures when other are characteristic the other will take the p with uh P which has more margin with others

the setup for the prayer is simply disre described the action space is defined in discrete space movement for five directions and weight weight current position totally six action is assumed a simple variational o encod like model is used to the internal model and the model is with the offline man the model is used to predict the expected free energy for each

action first the evaluation result when there is only one other besides the prayer as shown we discussed the poity of the agent based on the for view point the first one if my travel my travel distance is longer it is social behavior and the second if the other travel distance is shorter it is social behavior the start if the minimum margin among agent is larger it is social behavior

for if the inequality of the travel distance is smaller it is social behavior inequality means that the difference in Trav distance among agents as a comparison we also show the result of the standard active inference cases the two figures in the left side are the case when the others are selfish

in this case the prayer with the standard active inference moves straight to the destination because the prayer Act optimally only for its own purpose on the other hand the proposed agent to Appo the others as well as as well as the other because the prayer predict that the other expect to the prayer to avoid collision and the prayer try to respond to it as a result of taking this the two P the sociality of the behavior is increased namely the travel distance of the prayer increased compared to the that of the standard active inference case at the same time the travel distance of the others decrease and the minimum distance between the prayer and the other increase in addition the inequality of the difference in travel distance is decreased next the case where other aristic is shown on the light two figures in this case the same tendency is observed as in the case where of others other is selfish but the player moved a pass with larger margin to the other to the other compared to the case of selfish other the result indicates that the proposed agent can behave in Social manner that is that beneficial to the other even if the result are disadvantageous to itself an important point is that the difference of the social behavior does not depend on any reward designing as in reinforcement running namely the sociality depend on the character of the

other around the

prayer the other around the prayer is

not just obstacle for the agent but uh

as a a teacher for the social

[Music]

manner

s in the case where three others besides

the prayer are similar

result observe in the aristic as a case

namely the travel distance of the prayer

is increased and that of others are

decreased and the minimum margin is

increased and inequality of the travel

distance is decreased on the other hand

in the case of selfish others a

collision occurred it may be due to high

density in the center point in this case

the proposed agent can avoid collision

by waiting the crossing of

others I repeat the important point

there is no need to design leward or

penalty for

sociality in each

situation a social behavior realized by

respon responding to the other

expectation finally I would like to

summarize the explanation of the paper

as I said at the first SL the motivation

of This research is to develop unified

mechanism for social

behavior the answer for the motivation

is that learning an internal model to

predict the future behavior of others

this is same as standard free energy

principle or inference more more over

predict others expectation about me with

the same internal model and try to

respond to the others

expectation this this process make a cyclic process and it encourage the agent to behave like surrounding others because due to mirror system other expectation to me are same same to my expectation to others this mechanism encourages agent to behave like surrounding

others this summary shows our idea for what is social behavior the answer is that social behavior is a behavior like surrounding others the others are not obstacle but teacher for social behavior this also mean that we are teacher of AI or robot for the future

Society that's so this is the

explanation of the

paper i' like to discuss about

it

thank

you would either of the other authors

like to say hello or to give any

remarks hello I

amaki also a researcher at and

um um I I have been studying on the

control of system especially system with

hardware and uh my recent interest is

how to apply the

after robot

system hello my name is H I'm

responsible for This research thank you

for giving us this wonderful

time thank

you okay many interesting um questions

but definitely very striking

results about the tolerance to a variety

of different social settings for the

active inference

algorithm

to by nature of always updating at each
time step based upon the
expectations able to kind of deal with a
range of different settings like
different numbers of Agents
and the different um Behavior types for
the

agents um I'll I'll ask some questions
that were submitted by hear you Sam
no no or okay now

okay

all good all good okay I will read some
questions that were submitted by email
by shohi wakayama and also we'll look to
the live chat so I'll start with the
fourth question he wrote I would like to
receive more detailed explanations of no
operation action the noop in general
because I noticed also in the graphics
that when there's going to be a
collision that the active inference
agent just stops so how did you model or
calibrate that decision to make a move
at all versus which way to move
forward

okay first I'd like to answer what is
the no operation a
NOP I mean no is just a no
operation

so

in the evaluation case not means wait
the current

po the agent take no movement action the
agent stay the current position this is
a not operation no
operation and the question is very
important

because this is a critical problem to realize our method the problem is that we want to decide an action but the procedure we proposed requires to decide action as described in this figure the let Circle action actions for others we decide to action of the agent but it requires the deciding other action this cyclic requirement is very difficult problem so we have to set some option for the others the S

I think there is there are solution straty three solution strategies the first one is assume random action apprising the result of the randomly selected actions the second is predict predict action learning and predict using policy Network for example Alpha go use this strategy and the last one is we selected the thir strategy is that assume specific action no operation means I don't take any actions and you act as a priority please I mean NOP I mean this idea you for assuming no so I believe that humans has such sociality in natur this is the reason why I use the sad idea does it make sense yes okay next um

question how do you tune or calibrate the model so that it's in an area of State space that's pro-social like maybe you could have a generative model that just only did nope and it was just paralyzed or another one just goes straightforward and no matter what so how do you explore that state space and choose which parameters are open for updating and which parameters are going to be like explored during the design phase and then fixed at runtime oh at runtime uh I don't adjust any parameter at long time the model is Tred and offline Manner and the parameter model is explore explore searched at the training phas but the parameter space is not so huge in this case because the model is very simple the observation is just a two dimensional information X and Y that means location of the agent so in the case of four agent there is just eight eight dimensional space but we need sometime steps so but even that case the dimension is not so large so the hidden state is also not so lar can you understand yes evaluation situation is very simple so the model is also simple

so I don't

search parameter with a rot of IND

[Music]

that is not

PA simply just to try to some parameter

and I pick up a best parameter

set so this was a this was an in

silico software only simulation right oh

yes this is just the simulation about

simulation so um how did you tune it and

then I guess a bigger question and

definitely one that a lot of people are

interested in in learning active

inference is like how do you go from

having a in silico

simulation to bringing that onto a

physical

robot like what are the steps in the

process between having the

situation in the simulation and bringing

it into like a

room yes that's a very interesting point

and it's very important and Han try

to app the active inference into the

real robot so how about do you think

kind

um so

uh so I I I I I also tried in a small uh

space for to apply the this after to the

robot so um I have just an idea that um

um maybe uh The Continuous space is the

important um key for to uh PRS

simulation to the physical

world

and in general uh the problem is also

the problem in reinforcement running I

think so the techniques in reinforcement

running field also can be applied into

the active inference cases for example
simulation to simulation to real I I
don't remember the name of technique
but s
maybe such techniques can be applied in
the case of active
inference but I don't know there is a
active inference specific
technique I don't know such
Tech that's interesting about the
continuous State spaces because in the
discrete State space where you have the
markov decision process you get the big
planning trees but our experience like
walking through a crowd or maybe waiting
at a intersection it's not necessarily
based upon planning it's a lot more on
the path and the curvature of the path
that's planned and maybe the paths that
others are expected to take as they move
through space to
so that's possibly like a simpler
representation yes that could give
natural movement Behavior with a just a
few

[Music]

parameters thank you and there is a
another problem for applying this
technique to a real world that is a
computational cost because we use the
monte carlo simulation to start the best
option I think this is not sweet to to
the real situation so if we try to apply
this technique to the real application
we have to solve the computational cost
problems I think
so yeah another part even outside of the
decision making

computational costs would be like going from the video camera data oh yes when you're getting the exact location data perfectly you're in the minimum you're like kind of playing chess in the state Space versus to have noisy sensors there might need to be a lot of um uh observation model here a lot of layers before the simple kind of active INF could even

happen okay I'll ask another question from uh shohi so question five he wrote in this study the behaviors of the standard active inference agent and the empathetic agent are compared I would like to know why you did not compare the proposed method with other game theoretic approaches because the metrics like travel distance can be applicable to these methods as well

oh the reason why I compare the proposed method with active inference is that the aim of the paper is to extend the active inference to the social social behavior manner so it is not I think it is natural to compare the proposed method with standard active

inference but the question is that why we don't compare the proposed method with the other game theoretic method I don't I'm not sure what this means the game theoretic method but if that means the reinforcement learning method

the reason why I don't compare it is that just it needs a lot of time to

compare it

and

I I think it is important but the main focus of this paper is to evaluate the

quantitative future of the propos method and the

qualitative variation

is not so

important for us at the

moment but I think it is important and I

have to compare it in future

work so there is

no real reason this is just

uh we don't have uh much time to compare

it thank you makes

sense um another question so the

movement was

being influenced by the relative ratio

of the F_{other} and F

goal yes

so here F_{other} is kind of like a um

electrostatic

repulsion from other

agents how does it how does it come into

play that the red agent is entertaining

beliefs about what other agents

think

sorry can you repeat the question yeah

so here the two forces acting on the red

agent are the the gravity for the goal

and the repulsion for the proximity of

other

agents yes

so where or how does it come into to

play that the agent is

actually thinking what other agents are

going to

do like the birds flocking
simulations that have the simple rules
like go in the same direction as the
other birds and maintain a distance
between them but that doesn't require a
theory of mind for the other
birds

Oh you mean how the agent know the H
from the others

or ah you mean how the agent know the
course of goal of

others yeah it's just running away from
others or is it actually imagining what
they

think

so uh that's good question yes

uh in this

simulation

uh

s the answer is that the

destination destination the others

uh

given I don't remember the detail of the
evaluation I

think

but a

Sor the agent don't have to know the
information at first I think I give the
information to the agent but
that

information does not

need the

[Music]

asan

yeah

so the goals of the others are not not
expressely given but uh the with
internal model uh the agent can

predict um the
path test so we can
pred uh go like go but uh not not
exped does it exp does the red agent
explicitly model what the others are
going to
do or it just has heris stics that it
follows when they
move the foral one so only predict
the action recent action
and yes near
future yes the agent just learn the
future position of the others this is
the all of
the task of the internal model
so there's no need to give the go of the
others yes I think this is very
interesting aspect like to what extent
the social behavior rests upon Simple
Rules like
etiquette like passing on the right or
on the left that don't
require knowledge about the internal
states of others or even what kind of
thing they are whereas the experience of
sociality and a lot of the psychology of
Sociology focuses on very cognitive
mechanisms that are
very
expressive
thank
you this is because the action is
repeated to decided in short time
duration
so this
is uh future of the proposed method as
you
said thank you I don't notice that that

is
important
um
where are you going to go in your
continuing research or development
directions
sorry future
[Music]
please
uh I
think uh an important point is the
regulating the
empathy and we have a presentation the
last year uh conference about artificial
life in the
presentation we
discussed how regulates the emotion our
empathy this mean that we to uh the
paper we in is that the expected the pre
energy is extended to the including
others so D called D my plus the others
and the there is there should be weight
of empathy that means
uh
W in this equation we have to regulate
this
parameter I
don't consider the parameter in the
Journal paper but we have to
regulate this parameter because
there's diverse others there's good
person or there's part person we should
not empathy to the part address so this
is the important point I'll emphasize
others and
we tackle this problem with the emotion
emo we tied this problem with
the regard to the

emotional

context okay this is a future work and we publish uh we have a presentation at the at the current conference so if you have some interesting please read this

paper that's very interesting like in a lot of the active inference and social sciences conversations that we've had here there's a lot of discussion about like how do you balance the the coefficients in the free energy calculations for self and others yes because you're making decisions at a very short time scale like speech and second by second but then sociality is very longterm and so you could make a decision that is um aligned locally but then very out of alignment at a longer time scale especially if these other four have complex relation relationships then action is not very simple

Yes is there any other um things that the other authors want to say or anything else you want to talk about

I I don't

[Music]

have cool well thank you very much for joining the discussion thank you thank you yes good luck with the ongoing work and I hope to stay in touch and hear more about the research in the

future thank very

much

okay thank you see you

later

bye

fine

for